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TRANSISTOR AND SEMICONDUCTOR DEVICE USING THE SAME

Abstract

A semiconductor device includes a first transistor and a second transistor. The first transistor includes a plurality of first active channels and a first metal gate. The first active channels are stacked to each other. The first metal gate is formed on the first active channels. The first metal gate includes a plurality of first P-metal layers and a first N-metal layer stacked to the first N-metal layer. The second transistor includes a plurality of second active channels and a second metal gate. The second active channels are stacked to each other. The second metal gate is formed on the second active channels, wherein the second metal gate includes a second N-metal layer.

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Background/Summary

BACKGROUND

[0001] Conventional BJT (bipolar junction transistor) device is commonly used for bandgap reference circuit. However, The BJT occupies a larger area and has a higher power-costing, and thus it is not competitive for shrinkage node.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. 1A illustrates a schematic diagram of a first transistor of a semiconductor structure according to an embodiment of the present disclosure;

[0004] FIG. 1B illustrates a schematic diagram of a second transistor of the semiconductor structure according to an embodiment of the present disclosure;

[0005] FIG. 1C illustrates a schematic diagram of a second transistor of the semiconductor structure according to another embodiment of the present disclosure;

[0006] FIG. 1D illustrates a schematic diagram of a first transistor of the semiconductor structure **100** according to another embodiment of the present disclosure;

[0007] FIG. 2A illustrates a schematic diagram of a first circuit of the semiconductor structure in an embodiment of the present disclosure;

[0008] FIG. 2B illustrates a schematic diagram of a second circuit of the semiconductor structure in another embodiment of the present disclosure;

[0009] FIG. 2C illustrates a schematic diagram of a third circuit of the semiconductor structure in another embodiment of the present disclosure;

[0010] FIG. 2D illustrates a schematic diagram of a fourth circuit of the semiconductor structure in another embodiment of the present disclosure;

[0011] FIG. 2E illustrates a schematic diagram of a fifth circuit of the semiconductor structure in another embodiment of the present disclosure;

[0012] FIG. 2F illustrates a schematic diagram of a sixth circuit of the semiconductor structure in another embodiment of the present disclosure;

[0013] FIG. 2G illustrates a schematic diagram of a seventh circuit of the semiconductor structure in another embodiment of the present disclosure;

[0014] FIG. 3 illustrates a schematic diagram of the current mirror circuit of the semiconductor structure in FIG. 2A.

[0015] FIG. 4 illustrates a relationship between the temperature and the reference voltage in an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0016] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and

second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0017] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0018] As illustrated in FIGS. 1A to 1D, FIG. 1A illustrates a schematic diagram of a first transistor **110** of a semiconductor structure **100** according to an embodiment of the present disclosure, FIG. 1B illustrates a schematic diagram of a second transistor **120** of the semiconductor structure **100** according to an embodiment of the present disclosure, FIG. 1C illustrates a schematic diagram of a second transistor **130** of the semiconductor structure **100** according to another embodiment of the present disclosure, and FIG. 1D illustrates a schematic diagram of a first transistor **140** of the semiconductor structure **100** according to another embodiment of the present disclosure.

[0019] As illustrated in FIGS. 1A to 1D, the semiconductor structure **100** includes a plurality of transistors, for example, at least one first transistor **110**, at least one second transistor **120**, at least one second transistor **130** and at least one first transistor **140**. Although not illustrated, the semiconductor structure **100** further includes a substrate, for example, a portion of silicon wafer. The transistors are formed on the substrate.

[0020] As illustrated in FIG. 1A, the first transistor **110** is, for example, a flip-gate NMOS (or called “FGD”). The first transistor **110** includes a plurality of first active channels **111** stacked to each other, a first metal gate **112**, a first epitaxy **113A**, a second epitaxy **113B**, a spacer **114**, an interlayer dielectric (ILD) layer **115**, a contact etch stop layer (CESL) **116** and at least one inner spacer SP.

[0021] As illustrated in FIG. 1A, the first active channel **111** is, for example, silicon sheet (for example, nano-sheet). The first epitaxy **113A** may be one of a source and a drain of the first transistor **110**, and the second epitaxy **113B** may be another of the source and the drain of the first transistor **110**. The spacer **114** is formed on a lateral surface of the CESL **116**. The CESL **116** surrounds the ILD layer **115**. The spacer **114**, the ILD layer **115** and the CESL **116** are formed above the first epitaxy **113A** and the second epitaxy **113B**. The inner spacer SP is formed on a lateral surface of a portion of the first metal gate **112** which is located between the first active channel **111**.

[0022] As illustrated in FIG. 1A, the first metal gate **112** is formed on and/or between the first active channels **111**. The first metal gate **112** includes a first N-metal layer **1121**, a plurality of first P-metal layers **1122A** and **1122B**, an interaction layer (IL) **1123**, a high-k layer **1124**, a capping layer **1125**, a barrier layer **1126**, a glue layer **1127** and a metal layer **1128**.

[0023] As illustrated in FIG. 1A, the interaction layer **1123**, the high-k layer **1124**, the capping layer **1125**, the barrier layer **1126**, the first P-metal layer **1122B**, the first P-metal layer **1122A**, the first N-metal layer **1121**, the glue layer **1127** and the metal layer **1128** are formed, in order, in a manufacturing process of the transistor.

[0024] In an embodiment, the first N-metal layer **1121** may be formed of a material including TAL material. The first P-metal layer **1122A** and the first P-metal layer **1122B** may be formed a material including, for example, TiN, etc. The interaction layer **1123** may be formed a material including, for example, TaN, etc. The high-k layer **1124** may be formed of a material including: (i) a high-k dielectric material, such as hafnium oxide (HfO₂), titanium oxide (TiO₂), hafnium zirconium oxide (HfZrO), tantalum oxide (Ta₂O₃), hafnium silicate (HfSiO₄),

zirconium oxide (ZrO.sub.2), and zirconium silicate (ZrSiO.sub.2), and (ii) a high-k dielectric material having oxides of lithium (Li), beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), scandium (Sc), yttrium (Y), zirconium (Zr), aluminum (Al), lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), lutetium (Lu), (iii) other suitable high-k dielectric materials, or (iv) a combination thereof. As used herein, the term “high-k” refers to a high dielectric constant. In the field of semiconductor device structures and manufacturing processes, high-k refers to a dielectric constant that is greater than the dielectric constant of SiO.sub.2 (e.g., greater than 3.9). The capping layer **1125** may be formed a material including, for example, TiN, etc. The barrier layer **1126** may be formed a material including, for example, TaN, etc. The glue layer **1127** may be formed a material including, for example, TiN, etc. The metal layer **1128** may be formed a material including, for example, tungsten, etc.

[0025] In addition, the P-metal layer has a work-function different from that of the N-metal layer, and one of the P-metal layers may have the work-function different from or the same as another of the P-metal layers.

[0026] As illustrated in FIG. **1B**, the second transistor **120** is, for example, a negative-gate device NMOS (or called “NGD”). The second transistor **120** includes a plurality of second active channels **121** stacked to each other, a second metal gate **122**, a first epitaxy **123A**, a second epitaxy **123B**, a spacer **124**, an ILD layer **125** and a CESL **126** and at least one inner spacer SP.

[0027] As illustrated in FIG. **1B**, the second active channel **121** is, for example, silicon sheet (for example, nano-sheet). The first epitaxy **123A** may be one of a source and a drain of the second transistor **120**, and the second epitaxy **123B** may be another of the source and the drain of the second transistor **120**. The spacer **124** is formed on a lateral surface of the CESL **126**. The CESL **126** surrounds the ILD layer **125**. The spacer **124**, the ILD layer **125** and the CESL **126** are formed above the first epitaxy **123A** and the second epitaxy **123B**. The inner spacer SP is formed on a lateral surface of a portion of the second metal gate **122** which is located between the second active channel **121**.

[0028] As illustrated in FIG. **1B**, the second metal gate **122** is formed on and between the second active channels **121**. The second metal gate **122** includes a second N-metal layer **1221**, an interaction layer **1223**, a high-k layer **1224**, a capping layer **1225**, a barrier layer **1226**, a glue layer **1227** and a metal layer **1228**. Difference from the first metal gate **112** is that the second metal gate **122** may omit the first P-metal layer. As a result, a second driving voltage threshold $V_{sub.t2}$ for the second metal gate **122** (or the second transistor **120**) is less than a first driving voltage threshold $V_{sub.t1}$ for the first metal gate **112** (or the first transistor **110**). In an embodiment, the second driving voltage threshold $V_{sub.t2}$ for the second metal gate **122** (or the second transistor **120**) is, for example, 0.38 volts, and the first driving voltage threshold $V_{sub.t1}$ for the first metal gate **112** (or the first transistor **110**) is, for example, 0.72 volts. The driving voltage threshold is the voltage which may successfully drive the transistor to work.

[0029] As illustrated in FIG. **1B**, the interaction layer **1223**, the high-k layer **1224**, the capping layer **1225**, the barrier layer **1226**, the second N-metal layer **1221**, the glue layer **1227** and the metal layer **1228** are formed, in order, in a manufacturing process of the transistor.

[0030] In an embodiment, the second N-metal layer **1221** may be formed of a material the same as or similar to that of the first N-metal layer **1121**. The interaction layer **1223** may be formed a material the same as or similar to that of the interaction layer **1123**. The high-k layer **1224** may be formed of a material the same as or similar to that of the high-k layer **1124**. The capping layer **1225** may be formed a material the same as or similar to that of the capping layer **1125**. The barrier layer **1226** may be formed a material the same as or similar to that of the barrier layer **1126**. The glue layer **1227** may be formed a material the same as or similar to that of the glue layer **1127**. The metal layer **1228** may be formed a material the same as or similar to that the metal layer **1128**.

[0031] As illustrated in FIG. **1C**, the second transistor **130** is, for example, a negative-gate device

NMOS (or called “NGD”). The second transistor **130** includes a plurality of second active channels **131** stacked to each other, a second metal gate **132**, a first epitaxy **133A**, a second epitaxy **133B**, a spacer **134**, an ILD layer **135** and a CESL **136**.

[0032] As illustrated in FIG. **1C**, the second active channel **131** is, for example, silicon sheet (for example, nano-sheet). The first epitaxy **13A** may be one of a source and a drain of the second transistor **130**, and the second epitaxy **133B** may be another of the source and the drain of the second transistor **130**. The spacer **134** is formed on a lateral surface of the CESL **136**. The CESL **136** surrounds the ILD layer **135**. The spacer **134**, the ILD layer **135** and the CESL **136** are formed above the first epitaxy **133A** and the second epitaxy **133B**. The inner spacer SP is formed on a lateral surface of a portion of the first metal gate **132** which is located between the first active channel **131**.

[0033] As illustrated in FIG. **1C**, the second metal gate **132** is formed on and between the second active channels **131**. The second metal gate **132** includes a second N-metal layer **1321**, a second P-metal layer **1322A**, an interaction layer **1323**, a high-k layer **1324**, a capping layer **1325**, a barrier layer **1326**, a glue layer **1327** and a metal layer **1328**. Difference from the second metal gate **122** is that the number of the second P-metal layer of the second metal gate **132** is greater than the number of the second P-metal layer of the second metal gate **122**. As a result, a second driving voltage threshold $V_{sub.t2}$ for the second metal gate **132** (or the second transistor **130**) is greater than the second driving voltage threshold $V_{sub.t2}$ for the second metal gate **122** (or the second transistor **120**). Similarly, the number of the second P-metal layer of the second metal gate **132** is less than the number of the first P-metal layer of the first metal gate **112**. As a result, the second driving voltage threshold $V_{sub.t2}$ for the second metal gate **132** (or the second transistor **130**) is less than the first driving voltage threshold $V_{sub.t1}$ for the first metal gate **112** (or the first transistor **110**). In an embodiment, the second driving voltage threshold for the second metal gate **132** (or the second transistor **130**) is, for example, 0.54 volts.

[0034] As illustrated in FIG. **1C**, the interaction layer **1323**, the high-k layer **1324**, the capping layer **1325**, the barrier layer **1326**, the second P-metal layer **1322A**, the second N-metal layer **1321**, the glue layer **1327** and the metal layer **1328** are formed, in order, in a manufacturing process of the transistor.

[0035] In an embodiment, the second N-metal layer **1321** may be formed of a material the same as or similar to that of the first N-metal layer **1121**. The second P-metal layer **1322A** may be formed of a material the same as or similar to that of the first P-metal layer **1122A**. The interaction layer **1323** may be formed a material the same as or similar to that of the interaction layer **1123**. The high-k layer **1324** may be formed of a material the same as or similar to that of the high-k layer **1124**. The capping layer **1325** may be formed a material the same as or similar to that of the capping layer **1125**. The barrier layer **1326** may be formed a material the same as or similar to that of the barrier layer **1126**. The glue layer **1327** may be formed a material the same as or similar to that of the glue layer **1127**. The metal layer **1328** may be formed a material the same as or similar to that the metal layer **1128**.

[0036] As illustrated in FIG. **1D**, the first transistor **140** is, for example, a flip-gate NMOS (or called “FGD”). The first transistor **140** includes a plurality of first active channels **141** stacked to each other, a first metal gate **142**, a first epitaxy **143A**, a second epitaxy **143B**, a spacer **144**, an ILD layer **145** and a CESL **146**.

[0037] As illustrated in FIG. **1D**, the first active channel **141** is, for example, silicon sheet (for example, nano-sheet). The first epitaxy **143A** may be one of a source and a drain of the first transistor **140**, and the second epitaxy **143B** may be another of the source and the drain of the first transistor **140**. The spacer **144** is formed on a lateral surface of the CESL **146**. The CESL **146** surrounds the ILD layer **145**. The spacer **144**, the ILD layer **145** and the CESL **146** are formed above the first epitaxy **143A** and the second epitaxy **143B**. The inner spacer SP is formed on a lateral surface of a portion of the first metal gate **142** which is located between the first active

channel **141**.

[0038] As illustrated in FIG. 1D, the first metal gate **142** is formed on and between the first active channels **141**. The first metal gate **142** includes a first N-metal layer **1421**, a plurality of first P-metal layers **1422A**, **1422B** and **1422C**, an interaction layer **1423**, a high-k layer **1424**, a capping layer **1425**, a barrier layer **1426**, a glue layer **1427** and a metal layer **1428**. Difference from the second metal gate **122** is that the number of the first P-metal layer of the first metal gate **142** is greater than the number of the second P-metal layer of the second metal gate **122**. As a result, a first driving voltage threshold $V_{sub.t1}$ for the first metal gate **142** (or the first transistor **140**) is greater than the second driving voltage threshold $V_{sub.t2}$ for the second metal gate **122** (or the second transistor **120**). In an embodiment, the first driving voltage threshold $V_{sub.t1}$ for the first metal gate **142** is, for example, 0.844 volts.

[0039] In another embodiment, the number of the P-metal layers may be greater than 3.

[0040] As described above, the driving voltage threshold of the metal gate increases with the increasing of the number of the P-metal layers.

[0041] As illustrated in FIG. 1D, the interaction layer **1423**, the high-k layer **1424**, the capping layer **1425**, the barrier layer **1426**, the first P-metal layer **1422C**, the first P-metal layer **1422B**, the first P-metal layer **1422A**, the first N-metal layer **1421**, the glue layer **1427** and the metal layer **1428** are formed, in order, in a manufacturing process of the transistor.

[0042] In an embodiment, the first N-metal layer **1421** may be formed of a material the same as or similar to that of the first N-metal layer **1121**. The first P-metal layer **1422A** may be formed of a material the same as or similar to that of the first P-metal layer **1122A**. The first P-metal layer **1422B** may be formed of a material the same as or similar to that of the first P-metal layer **1122B**. The first P-metal layer **1422C** may be formed of a material including, for example, TiN, etc. The interaction layer **1423** may be formed a material the same as or similar to that of the interaction layer **1123**. The high-k layer **1424** may be formed of a material the same as or similar to that of the high-k layer **1124**. The capping layer **1425** may be formed a material the same as or similar to that of the capping layer **1125**. The barrier layer **1426** may be formed a material the same as or similar to that of the barrier layer **1126**. The glue layer **1427** may be formed a material the same as or similar to that of the glue layer **1127**. The metal layer **1428** may be formed a material the same as or similar to that the metal layer **1128**.

[0043] As illustrated in FIGS. 2A to 2G, FIG. 2A illustrates a schematic diagram of a first circuit **10** of the semiconductor structure **100** in an embodiment of the present disclosure, FIG. 2B illustrates a schematic diagram of a second circuit **20** of the semiconductor structure **100** in another embodiment of the present disclosure, FIG. 2C illustrates a schematic diagram of a third circuit **30** of the semiconductor structure **100** in another embodiment of the present disclosure, FIG. 2D illustrates a schematic diagram of a fourth circuit **40** of the semiconductor structure **100** in another embodiment of the present disclosure, FIG. 2E illustrates a schematic diagram of a fifth circuit **50** of the semiconductor structure **100** in another embodiment of the present disclosure, FIG. 2F illustrates a schematic diagram of a sixth circuit **60** of the semiconductor structure **100** in another embodiment of the present disclosure, and FIG. 2G illustrates a schematic diagram of a seventh circuit **70** of the semiconductor structure **100** in another embodiment of the present disclosure.

[0044] As illustrated in FIG. 2A, the first transistor **110** is located at a first region **A1**, and the second transistor **130** is located at a second region **A2**. In an embodiment, all transistors in the first region **A1** are FGD, while all transistors in the second region **A2** are NGD. In another embodiment, all transistors in the first region **A1** may be a combination of the FGD and the NGD, and/or all transistors in the second region **A2** may be a combination of the FGD and the NGD.

[0045] As illustrated in FIG. 2A, the first epitaxy **113A** may be the source of the first transistor **110**, and the second epitaxy **113B** may be the drain of the first transistor **110**. The first epitaxy **133A** may be the source of the second transistor **130**, and the second epitaxy **133B** may be the drain of the second transistor **130**. The first metal gate **112** of the first transistor **110** is electrically

connected with the second metal gate **132** of the second transistor **130**, and the first epitaxy **113A** of the first transistor **110** is electrically connected with a first node **151** of a current mirror circuit **150**. The current mirror circuit **150** may supply a driving current (or subthreshold current of the FGD) **I** to the first transistor **110**. The first epitaxy **133A** of the second transistor **130** is electrically connected with a second node **152** of the current mirror circuit **150**. The current mirror circuit **150** electrically connects a device voltage **V.sub.DD** with the first epitaxy **113A** of the first transistor **110**. The current mirror circuit **150** is configured to convert the device voltage **V.sub.DD** to a node voltage **V_a**.

[0046] As illustrated in FIG. 2A, a first resistor **R1** electrically connects a third node **153** of the current mirror circuit **150** with a series voltage **V.sub.SS**, and a second resistor **R2** electrically connects the second epitaxy **133B** of the second transistor **130** with the series voltage **V.sub.SS**. A reference node **REF** between the current mirror circuit **150** and the first resistor **R1** has a reference voltage **V.sub.ref**. In an embodiment, the first resistor **R1** has a resistance greater than that of the second resistor **R2**, and thus it may amplify the reference voltage **V.sub.ref**. The reference voltage **V.sub.ref** is proportional to a difference between the first driving voltage threshold **V.sub.t1** and the second driving voltage threshold **V.sub.t2**.

[0047] As illustrated in FIG. 2A, the semiconductor structure **100** further includes a fifth transistor **160**. The fifth transistor **160** electrically connects a line between the second epitaxy **133B** of the second transistor **130** and the second resistor **R2** with the second epitaxy **113B** of the first transistor **110**. The fifth transistor **160** is configured to conduct a leakage current to a grounding.

[0048] In an embodiment, as shown in Table 1, in case of the first region **A1** including one first transistor **110** and the second region **A2** including one second transistor **130**, the reference voltage **V.sub.ref** is about $0.2V \pm 0.02 V$. In case of the first region **A1** including one first transistor **140** and the second region **A2** including one second transistor **130**, the reference voltage **V.sub.ref** is about $0.3V \pm 0.02 V$. In case of the first region **A1** including one first transistor **110** and the second region **A2** including one second transistor **120**, the reference voltage **V.sub.ref** is about $0.35V \pm 0.02 V$. In case of the first region **A1** including one first transistor **140** and the second region **A2** including one second transistor **120** (that is, the circuit in FIG. 2D), the reference voltage **V.sub.ref** is about $0.5V \pm 0.02 V$. Furthermore, due to the semiconductor structure **100** including the FGD transistor (for example, the first transistor **110** and/or the first transistor **140**), and it may increase the reference voltage **V.sub.ref**. In addition, the more the number of the P-type metals of the FGD transistor in the first region **A1** is, the greater the reference voltage **V.sub.ref** is.

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TABLE 1	first region A1	first transistor 110	first transistor 140	second region A2 (FGD)	(FGD) second transistor 130 (NGD)	$0.2 V \pm 0.02 V$	$0.3 V \pm 0.02 V$	second transistor 120 (NGD)	$0.35 V \pm 0.02 V$	$0.5 V \pm 0.02 V$
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[0049] As illustrated in FIG. 2B, two first transistors **110** connected with each other are located at the first region **A1**, and two second transistors **130** connected with each other are located at the second region **A2**. The first metal gate **112** of one of the two first transistors **110** is electrically connected with a line **L11** between the two first transistors **110**. The second metal gate **132** of one of the two second transistors **130** is electrically connected with a line **L21** between the two second transistors **130**.

[0050] As illustrated in FIG. 2B, the first epitaxy **113A** may be the source of the first transistor **110**, and the second epitaxy **113B** may be the drain of the first transistor **110**. The first epitaxy **113A** of the first transistor **110** is electrically connected with the first node **151** of the current mirror circuit **150**. The current mirror circuit **150** may supply the driving current **I** to the first transistor **110**. The first metal gate **112** of a first one (the upper one) of the first transistors **110** is electrically connected with the second metal gate **132** of a first one (the upper one) of the second transistors **130**, the first epitaxy **113A** of a second one (the lower one) of the first transistors **110** is electrically connected with the second epitaxy **113B** of the first one of the first transistors **110**, and the second epitaxy **113B** of the second one of the first transistors **110** is electrically connected with the series voltage

V.sub.SS.

[0051] As illustrated in FIG. 2B, the first epitaxy **133A** may be the source of the second transistor **130**, and the second epitaxy **133B** may be the drain of the second transistor **130**. The first epitaxy **133A** of a first one (the upper one) of the second transistors **130** is electrically connected with the second node **152** of the current mirror circuit **150**. The first epitaxy **133A** of a second one (the lower one) of the second transistors **130** is electrically connected with the second epitaxy **133B** of the first one (the upper one) of the second transistors **130**, and the second epitaxy **133B** of the second one of the second transistors **130** is electrically connected with the second resistor **R2**.

[0052] As illustrated in FIG. 2B, the first resistor **R1** electrically connects the third node **153** of the current mirror circuit **150** with the series voltage V.sub.SS, and the second resistor **R2** electrically connects the second epitaxy **133B** of the second one of the second transistors **130** with the series voltage V.sub.SS. The reference node REF between the current mirror circuit **150** and the first resistor **R1** has the reference voltage V.sub.ref. In an embodiment, the first resistor **R1** has the resistance greater than that of the second resistor **R2**, and thus it may amplify the reference voltage V.sub.ref. The fifth transistor **160** electrically connects a line between the second epitaxy **133B** of the second one of the second transistors **130** and the second resistor **R2** with the second epitaxy **113B** of the second one of the first transistors **110**. The fifth transistor **160** may conduct the leakage current to the grounding.

[0053] In another embodiment, the second transistors **130** in FIG. 2B may be replaced by the second transistors **120** and/or the first transistors **110** in FIG. 2B may be replaced by the first transistors **140**.

[0054] As illustrated in FIG. 2C, three first transistors **110** connected with each other are located at the first region **A1**, and three second transistors **130** connected with each other are located at the second region **A2**. The first metal gate **112** of a first one (the middle one) of the three first transistors **110** is electrically connected with a first line **L11** between the first one and a second one (the upper one) of the three first transistors **110**. The first metal gate **112** of a third one (the lower one) of the three first transistors **110** is electrically connected with a second line **L2** between the first one and the third one. The second metal gate **132** of a first one (the middle one) of the three second transistors **130** is electrically connected with a first line **L21** between the first one and a second one (the upper one) of the three second transistors **130**, the second metal gate **132** of a third one (the lower one) of the three second transistors **130** is electrically connected with a second line **L22** between the first one and the third one of the three second transistors **130**.

[0055] As illustrated in FIG. 2C, the first epitaxy **113A** may be the source of the first transistor **110**, and the second epitaxy **113B** may be the drain of the first transistor **110**. The first epitaxy **113A** of the second one (the upper one) of the first transistors **110** is electrically connected with the first node **151** of the current mirror circuit **150**. The current mirror circuit **150** may supply the driving current **I** to the first transistor **110**. The first metal gate **112** of the second one (the upper one) of the first transistors **110** is electrically connected with the second metal gate **132** of the second one (the upper one) of the second transistors **130**, the first epitaxy **113A** of the first one (the middle one) of the first transistors **110** is electrically connected with the second epitaxy **113B** of the second one (the upper one) of the first transistors **110**, and the second epitaxy **113B** of the first one (the middle one) of the first transistors **110** is electrically connected with the first epitaxy **113A** of the third one (the lower one) of the first transistors **110**. The second epitaxy **113B** of the third one of the first transistors **110** is electrically connected with the series voltage V.sub.SS.

[0056] As illustrated in FIG. 2C, the second epitaxy **133A** may be the source of the second transistor **130**, and the second epitaxy **133B** may be the drain of the second transistor **130**. The first epitaxy **133A** of the second one (the upper one) of the second transistors **130** is electrically connected with the second node **152** of the current mirror circuit **150**. The second metal gate **132** of the second one (the upper one) of the second transistors **130** is electrically connected with the first metal gate **112** of the second one (the upper one) of the first transistors **110**, the first epitaxy **133A**

of the first one (the middle one) of the second transistors **130** is electrically connected with the second epitaxy **133B** of the second one (the upper one) of the second transistors **130**, and the second epitaxy **133B** of the first one (the middle one) of the second transistors **130** is electrically connected with the first epitaxy **133A** of the third one (the lower one) of the second transistors **130**, and the second epitaxy **133B** of the third one (the lower one) of the second transistors **130** is electrically connected with the second resistor **R2**.

[0057] As illustrated in FIG. 2C, the first resistor **R1** electrically connects the third node **153** of the current mirror circuit **150** with the series voltage $V_{sub,SS}$, and the second resistor **R2** electrically connects the second epitaxy **133B** of the third one (the lower one) of the second transistors **130** with the series voltage $V_{sub,SS}$. The reference node REF between the third node **153** of the current mirror circuit **150** and the first resistor **R1** has the reference voltage $V_{sub,ref}$. In an embodiment, the first resistor **R1** has the resistance greater than that of the second resistor **R2**, and thus it may amplify the reference voltage $V_{sub,ref}$. The fifth transistor **160** electrically connects a line between the second epitaxy **133B** of the third one (the lower one) of the second transistors **130** and the second resistor **R2** with the second epitaxy **113B** of the third one (the lower one) of the first transistors **110**. The fifth transistor **160** may conduct the leakage current to the grounding.

[0058] As shown in Table 2 below, the more the number of the FD transistors is, the greater the reference voltage $V_{sub,ref}$ is, the less the range (a difference of the maximum reference voltage and the minimum reference voltage is, and the less the standard deviation of the reference voltages $V_{sub,ref}$ is. In addition, as long as the reference voltage $V_{sub,ref}$ in Table 2 may be obtained, the embodiment of the present disclosure does not limit the value of the first resistor **R1**, the value of the current **I**, the value of the second resistor **R2**, the value of the first driving voltage threshold $V_{sub,t1}$, the value of the second driving voltage threshold $V_{sub,t2}$ and/or the value of other relevant parameter.

TABLE-US-00002 TABLE 2 standard range deviation reference voltage $V_{sub,ref}$ Monte-carlo 1000 cycles Circuit 10 0.2 V $\pm 2.5\%$ <1% Circuit 20 0.35 V $\pm 1.5\%$ <0.5% Circuit 30 0.5 V $\pm 1.1\%$ <0.5%

[0059] As illustrated in FIG. 2D, the first transistor **140** is located at the first region **A1**, and the second transistor **120** is located at the second region **A2**. The first epitaxy **143A** may be the source of the first transistor **140**, and the second epitaxy **143B** may be the drain of the first transistor **140**. The first epitaxy **123A** may be the source of the second transistor **120**, and the second epitaxy **123B** may be the drain of the second transistor **120**. The first metal gate **142** of the first transistor **140** is electrically connected with the second metal gate **122** of the second transistor **120**, and the first epitaxy **143A** of the first transistor **140** is electrically connected with the first node **151** of the current mirror circuit **150**. The current mirror circuit **150** may supply the driving current **I** to the first transistor **140**. The first epitaxy **123A** of the second transistor **120** is electrically connected with the second node **152** of the current mirror circuit **150**. The current mirror circuit **150** electrically connects the device voltage $V_{sub,DD}$ with the first epitaxy **143A** of the first transistor **110**. The current mirror circuit **150** is configured to convert the device voltage $V_{sub,DD}$ to the node voltage V_a .

[0060] As illustrated in FIG. 2D, the first resistor **R1** electrically connects the third node **153** of the current mirror circuit **150** with the series voltage $V_{sub,SS}$, and the second resistor **R2** electrically connects the second epitaxy **123B** of the second transistor **120** with the series voltage $V_{sub,SS}$. The reference node REF between the current mirror circuit **150** and the first resistor **R1** has the reference voltage $V_{sub,ref}$. In an embodiment, the first resistor **R1** has the resistance greater than that of the second resistor **R2**, and thus it may amplify the reference voltage $V_{sub,ref}$.

[0061] As illustrated in FIG. 2D, the fifth transistor **160** electrically connects the line between the second epitaxy **123B** of the second transistor **120** and the second resistor **R2** with the second epitaxy **143B** of the first transistor **140**. The fifth transistor **160** is configured to conduct a leakage current to the grounding.

[0062] As illustrated in FIG. 2E, two first transistors **140** connected with each other are located at the first region **A1**, and two second transistors **120** connected with each other are located at the second region **A2**. The first metal gate **142** of one of the two first transistors **140** is electrically connected with the line **L11** between the two first transistors **140**. The second metal gate **122** of one of the two second transistors **120** is electrically connected with the line **L21** between the two second transistors **120**.

[0063] As illustrated in FIG. 2E, the first epitaxy **143A** may be the source of the first transistor **140**, and the second epitaxy **143B** may be the drain of the first transistor **140**. The first epitaxy **143A** of the first transistor **140** is electrically connected with the first node **151** of the current mirror circuit **150**. The current mirror circuit **150** may supply the driving current I to the first transistor **140**. The first metal gate **142** of a first one (the upper one) of the first transistors **140** is electrically connected with the second metal gate **122** of the first one (the upper one) of the second transistors **120**, the first epitaxy **143A** of the second one (the lower one) of the first transistors **140** is electrically connected with the second epitaxy **143B** of the first one (the upper one) of the first transistors **140**, and the second epitaxy **143B** of the second one (the lower one) of the first transistors **140** is electrically connected with the series voltage $V_{sub,SS}$.

[0064] As illustrated in FIG. 2E, the first epitaxy **123A** may be the source of the second transistor **120**, and the second epitaxy **123B** may be the drain of the second transistor **120**. The first epitaxy **123A** of a first one (the upper one) of the second transistor **120** is electrically connected with the second node **152** of the current mirror circuit **150**. The first epitaxy **123A** of the second one (the lower one) of the second transistors **120** is electrically connected with the second epitaxy **123B** of the first one (the upper one) of the second transistors **120**, and the second epitaxy **123B** of the second one (the lower one) of the second transistors **120** is electrically connected with the second resistor **R2**.

[0065] As illustrated in FIG. 2E, the first resistor **R1** electrically connects the third node **153** of the current mirror circuit **150** with the series voltage $V_{sub,SS}$, and the second resistor **R2** electrically connects the second epitaxy **123B** of the second one of the second transistors **120** with the series voltage $V_{sub,SS}$. The reference node **REF** between the current mirror circuit **150** and the first resistor **R1** has the reference voltage $V_{sub,ref}$. In an embodiment, the first resistor **R1** has the resistance greater than that of the second resistor **R2**, and thus it may amplify the reference voltage $V_{sub,ref}$. The fifth transistor **160** electrically connects the line between the second epitaxy **123B** of the second one (the lower one) of the second transistors **120** and the second resistor **R2** with the second epitaxy **143B** of the second one (the lower one) of the first transistors **140**. The fifth transistor **160** may conduct the leakage current to the grounding.

[0066] As illustrated in FIG. 2F, three first transistors **140** connected with each other are located at the first region **A1**, and three second transistors **120** connected with each other are located at the second region **A2**. The first metal gate **142** of the first one (the middle one) of the three first transistors **140** is electrically connected with the first line **L11** between the first one and a second one (the upper one) of the three first transistors **140**. The first metal gate **142** of the third one (the lower one) of the three first transistors **140** is electrically connected with the second line **L12** between the first one and the third one. The second metal gate **122** of the first one (the middle one) of the three second transistors **120** is electrically connected with the first line **L21** between the first one and the second one (the upper one) of the three second transistors **120**, the second metal gate **122** of the third one (the lower one) of the three second transistors **120** is electrically connected with the second line **L22** between the first one and the third one of the three second transistors **120**.

[0067] As illustrated in FIG. 2F, the first epitaxy **143A** may be the source of the first transistor **140**, and the second epitaxy **143B** may be the drain of the first transistor **140**. The first epitaxy **143A** of the second one (the upper one) of the first transistors **140** is electrically connected with the first node **151** of the current mirror circuit **150**. The current mirror circuit **150** may supply the driving current I to the first transistor **140**. The first metal gate **142** of the second one (the upper one) of the

first transistors **140** is electrically connected with the second metal gate **122** of the second one (the upper one) of the second transistors **120**, the first epitaxy **143A** of the first one (the middle one) of the first transistors **140** is electrically connected with the second epitaxy **143B** of the second one (the upper one) of the first transistors **140**, and the second epitaxy **143B** of the first one (the middle one) of the first transistors **140** is electrically connected with the first epitaxy **143A** of the third one (the lower one) of the first transistors **140**. The second epitaxy **143B** of the third one of the first transistors **140** is electrically connected with the series voltage $V_{sub,SS}$.

[0068] As illustrated in FIG. 2F, the second epitaxy **123A** may be the source of the second transistor **120**, and the second epitaxy **123B** may be the drain of the second transistor **120**. The first epitaxy **123A** of the second one (the upper one) of the second transistors **120** is electrically connected with the second node **152** of the current mirror circuit **150**. The second metal gate **122** of the second one (the upper one) of the second transistors **120** is electrically connected with the first metal gate **142** of the second one (the upper one) of the first transistors **140**, the first epitaxy **123A** of the first one (the middle one) of the second transistors **120** is electrically connected with the second epitaxy **123B** of the second one (the upper one) of the second transistors **120**, and the second epitaxy **123B** of the first one (the middle one) of the second transistors **120** is electrically connected with the first epitaxy **123A** of the third one (the lower one) of the second transistors **120**, and the second epitaxy **123B** of the third one (the lower one) of the second transistors **120** is electrically connected with the second resistor **R2**.

[0069] As illustrated in FIG. 2F, the first resistor **R1** electrically connects the third node **153** of the current mirror circuit **150** with the series voltage $V_{sub,SS}$, and the second resistor **R2** electrically connects the second epitaxy **123B** of the third one (the lower one) of the second transistors **120** with the series voltage $V_{sub,SS}$. The reference node **REF** between the third node **153** of the current mirror circuit **150** and the first resistor **R1** has the reference voltage $V_{sub,ref}$. In an embodiment, the first resistor **R1** has the resistance greater than that of the second resistor **R2**, and thus it may amplify the reference voltage $V_{sub,ref}$. The fifth transistor **160** electrically connects the line between the second epitaxy **123B** of the third one (the lower one) of the second transistors **120** and the second resistor **R2** with the second epitaxy **143B** of the third one (the lower one) of the first transistors **140**. The fifth transistor **160** may conduct the leakage current to the grounding.

[0070] As illustrated in FIG. 2G, the seventh circuit **70** includes the features similar to that of the sixth circuit **60**, and the difference is that a combination of at least one second transistor **120** and at least one second transistor **130** are disposed in the second region **A2**. In another embodiment, a combination of at least one first transistor **110** and at least one first transistor **140** are disposed in the first region **A1** of the seventh circuit **70**.

[0071] As shown in Table 3 below, the more the number of the FGD transistors is, the greater the reference voltage $V_{sub,ref}$ is, the less the range (the difference of the maximum reference voltage and the minimum reference voltage is, and the less the standard deviation of the reference voltages $V_{sub,ref}$ is. In addition, as long as the reference voltage $V_{sub,ref}$ in Table 3 may be obtained, the embodiment of the present disclosure does not limit the value of the first resistor **R1**, the value of the current **I**, the value of the second resistor **R2**, the value of the first driving voltage threshold $V_{sub,t1}$, the value of the second driving voltage threshold $V_{sub,t2}$ and/or the value of other relevant parameter.

TABLE-US-00003 TABLE 3 standard range deviation reference voltage $V_{sub,ref}$ Monte-carlo
1000 cycles Circuit 40 0.5 V $\pm 2.5\%$ <1% Circuit 50 1 V $\pm 1.5\%$ <0.5% Circuit 60 1.4 V $\pm 1.1\%$
<0.5% Circuit 70 1.2 V $\pm 1.1\%$ <0.5%

[0072] As described above, in an embodiment, the first region **A1** may include at least one FGD transistor (for example, the first transistor **110** and/or the first transistor **140**), and the second region **A2** may include at least one NGD transistor (for example, the second transistor **120** and/or the second transistor **130**). In another embodiment, the first region **A1** may include a combination of at least one FGD transistor (for example, the first transistor **110** and/or the first transistor **140**) and at

least one NGD transistor (for example, the second transistor **120** and/or the second transistor **130**), and/or the second region **A2** may include a combination of at least one FGD transistor (for example, the first transistor **110** and/or the first transistor **140**) and at least one NGD transistor (for example, the second transistor **120** and/or the second transistor **130**).

[0073] As illustrated in FIG. 3, FIG. 3 illustrates a schematic diagram of the current mirror circuit **150** of the semiconductor structure **100** in FIG. 2A. The current mirror circuit **150** includes a PMOS upper current mirror **150A**, a PMOS lower current mirror **150B** and a NMOS current mirror **150C** which are electrically connected with each other. Through a circuit design of the PMOS upper current mirror **150A**, the PMOS lower current mirror **150B** and the NMOS current mirror **150C**, the current mirror circuit **150** may supply the driving current I to the first transistor (for example, the first transistors **110** and **140**).

[0074] As illustrated in FIG. 4, FIG. 4 illustrates a relationship SI between the temperature and the reference voltage $V_{sub.ref}$ in an embodiment of the present disclosure. As illustrated in FIGS. 2A to 2G, the reference current $I_{sub.ref}$ (or subthreshold current of NGD) of the reference node REF is equal to I/n .

[0075] The reference voltage $V_{sub.ref}$ may be obtained by the formula (1) below, wherein $V_{sub.t1}$ is the first driving voltage threshold, $V_{sub.t2}$ is the second driving voltage threshold, T is temperature, and q is magnitude of the electrical charge on the electron, k is Boltzmann constant, and γ is sub-threshold swing coefficient. The sub-threshold swing coefficient γ depends on the gate oxide capacitance and/or the depletion layer capacitance of the source/drain junction of the semiconductor device **100**.

$$[00001] V_{ref} = (V_{t1} - V_{t2}) + \frac{1}{n} \times \ln(n) \times \frac{kT}{q} \quad (1)$$

[0076] The differentiation of the reference voltage $V_{sub.ref}$ over temperature (namely, $\partial V_{sub.ref} / \partial T$) may be obtained by the formula (2) below. When n is equal to 1, $\partial V_{sub.ref} / \partial T$ is equal to 0, and it means that the reference voltage $V_{sub.ref}$ does not change with temperature. The suitable or expected value of n may be obtained through adjusting the value of n in a simulation process or a testing process, and then the current mirror circuit **150** and/or other related circuits may be design according to the obtained value of n for making the reference voltage $V_{sub.ref}$ remain constant over temperature, or making the change of the reference voltage $V_{sub.ref}$ over temperature be less than a default value, as illustrated in FIG. 4. In an embodiment, the value of n may range between 0.01 and 100.

$$[00002] \frac{\partial V_{ref}}{\partial T} = \frac{\partial (V_{t1} - V_{t2})}{\partial T} + \frac{1}{n} \times \frac{kT}{q} \times \ln(n) \quad (2)$$

[0077] The above description of illustrated implementations of the disclosure, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. While specific implementations of, and examples for, the disclosure are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the disclosure, as those skilled in the relevant art will recognize.

[0078] These modifications may be made to the disclosure in light of the above detailed description. The terms used in the following claims should not be construed to limit the disclosure to the specific implementations disclosed in the specification and the claims. Rather, the scope of the disclosure is to be determined entirely by the following claims, which are to be construed in accordance with established doctrines of claim interpretation.

[0079] According to the present disclosure, a semiconductor device includes at least one FGD transistor and at least one NGD transistor connected with the FGD transistor. In comparison with the conventional semiconductor device (for example, BJT) which does not include the FGD transistor, the reference voltage range of the semiconductor device may meet general specification $\pm 3\%$ or greater. In addition, in comparison with the conventional semiconductor device which does not include the FGD transistor, the power saving for the semiconductor device is about 80%, and the area reduction for the semiconductor device is about 70%.

[0080] Example embodiment 1: a transistor includes a plurality of active channels and a metal gate. The active channels are stacked to each other. The metal gate is formed on the active channels, wherein the metal gate includes a plurality of P-metal layers and a N-metal layer stacked on the P-metal layers.

[0081] Example embodiment 2 based on Example embodiment 1: each of the P-metal layers is formed of TiN.

[0082] Example embodiment 3 based on Example embodiment 1: the metal gate further includes an interaction layer, a high-k layer, a capping layer and a barrier layer. The high-k layer is stacked on the interaction layer. The capping layer is stacked on the high-k layer. The barrier layer is stacked on the capping layer. The P-metal layers are stacked on the barrier layer.

[0083] Example embodiment 4 based on Example embodiment 3: the capping layer is formed of TiN.

[0084] Example embodiment 5 based on Example embodiment 3: the barrier layer is formed of TaN.

[0085] Example embodiment 6 based on Example embodiment 1: the metal gate further includes a metal layer and a glue layer. The glue layer is formed between the metal layer and the N-metal layer.

[0086] Example embodiment 7 based on Example embodiment 1: the number of the P-metal layers is equal to or greater than 3.

[0087] Example embodiment 8: a semiconductor device includes a first transistor and a second transistor. The first transistor includes a plurality of first active channels and a first metal gate. The first active channels are stacked to each other. The first metal gate is formed on the first active channels. The first metal gate includes a plurality of first P-metal layers and a first N-metal layer stacked to the first P-metal layers. The second transistor includes a plurality of second active channels and a second metal gate. The second active channels are stacked to each other. The second metal gate is formed on the second active channels, wherein the second metal gate includes a second N-metal layer.

[0088] Example embodiment 9 based on Example embodiment 8: the first transistor has a first driving voltage threshold, and the second transistor has a second driving voltage threshold less than the first driving voltage threshold.

[0089] Example embodiment 10 based on Example embodiment 8: the first metal gate of the first transistor is electrically connected with the second metal gate of the second transistor.

[0090] Example embodiment 11 based on Example embodiment 8: the number of the first P-metal layers is two or three.

[0091] Example embodiment 12 based on Example embodiment 8: the semiconductor device further includes two first transistors connected with each other and two second transistors connected with each other, and the first metal gate of one of the two first transistors is electrically connected with the second metal gate of one of the two second transistors.

[0092] Example embodiment 13 based on Example embodiment 8: the second metal gate further includes a second P-metal layer stacked to the second N-metal layer.

[0093] Example embodiment 14 based on Example embodiment 13: the semiconductor device further includes two first transistors connected with each other and two second transistors connected with each other, and the first metal gate of one of the two first transistors is electrically connected with the second metal gate of one of the two second transistors.

[0094] Example embodiment 15 based on Example embodiment 8: the semiconductor device further includes three first transistors connected with each other and three second transistors connected with each other, and the first metal gate of one of three first transistors is electrically connected with the second metal gate of one of the three second transistors.

[0095] Example embodiment 16 based on Example embodiment 8: the second metal gate further includes a second P-metal layer stacked to the second N-metal layer. The semiconductor device

further includes three first transistors connected with each other and three second transistors connected with each other, and the first metal gate of one of three first transistors is electrically connected with the second metal gate of one of the three second transistors.

[0096] Example embodiment 17 based on Example embodiment 8: the semiconductor device further includes a current mirror circuit, a first resistor and a second resistor. The current mirror circuit is electrically connected with the first transistor. The first resistor electrically connects the current mirror circuit with a series voltage. The second resistor electrically connects the second transistor with the series voltage.

[0097] Example embodiment 18: a semiconductor device includes a first transistor and a plurality of second transistors. The first transistor includes a plurality of first active channels and a first metal gate. The first active channels are stacked to each other. The first metal gate is formed on the first active channels. The first metal gate includes a plurality of first P-metal layers and a first N-metal layer stacked to the first P-metal layers. Each second transistor includes a plurality of second active channels and a second metal gate. The second active channels are stacked to each other. The second metal gate is formed on the second active channels, wherein the second metal gate includes a second N-metal layer. Wherein one of the second transistors further includes a second P-metal layer, and is electrically connected with another of the second transistors.

[0098] Example embodiment 19 based on Example embodiment 18: the first transistor has a first driving voltage threshold, and each second transistor has a second driving voltage threshold less than the first driving voltage threshold.

[0099] Example embodiment 20 based on Example embodiment 18: the semiconductor device further includes two first transistors connected with each other, and the first metal gate of one of the two first transistors is electrically connected with the second metal gate of one of the two second transistors.

[0100] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A transistor, comprising: a plurality of active channels stacked to each other; and a metal gate formed on the active channels, wherein the metal gate comprises: a plurality of P-metal layers; and a N-metal layer stacked on the P-metal layers.
2. The transistor as claimed in claim 1, wherein each of the P-metal layers is formed of TiN.
3. The transistor as claimed in claim 1, wherein the metal gate further comprises: an interaction layer; a high-k layer stacked on the interaction layer; a capping layer stacked on the high-k layer; and a barrier layer stacked on the capping layer; wherein the P-metal layers are stacked on the barrier layer.
4. The transistor as claimed in claim 3, wherein the capping layer is formed of TiN.
5. The transistor as claimed in claim 3, wherein the barrier layer is formed of TaN.
6. The transistor as claimed in claim 1, wherein the metal gate further comprises: a metal layer; and a glue layer formed between the metal layer and the N-metal layer.
7. The transistor as claimed in claim 1, wherein the number of the P-metal layers is equal to or greater than 3.
8. A semiconductor device, comprising: a first transistor, comprising: a plurality of first active

channels stacked to each other; and a first metal gate formed on the first active channels, wherein the first metal gate comprises: a plurality of first P-metal layers; and a first N-metal layer stacked to the first P-metal layers; a second transistor, comprising: a plurality of second active channels stacked to each other; and a second metal gate formed on the second active channels, wherein the second metal gate comprises a second N-metal layer.

9. The semiconductor device as claimed in claim 8, wherein the first transistor has a first driving voltage threshold, and the second transistor has a second driving voltage threshold less than the first driving voltage threshold.

10. The semiconductor device as claimed in claim 8, wherein the first metal gate of the first transistor is electrically connected with the second metal gate of the second transistor.

11. The semiconductor device as claimed in claim 8, wherein the number of the first P-metal layers is two or three.

12. The semiconductor device as claimed in claim 8, further comprising two first transistors connected with each other and two second transistors connected with each other, and the first metal gate of one of the two first transistors is electrically connected with the second metal gate of one of the two second transistors.

13. The semiconductor device as claimed in claim 8, wherein the second metal gate further comprises a second P-metal layer stacked to the second N-metal layer.

14. The semiconductor device as claimed in claim 13, further comprising two first transistors connected with each other and two second transistors connected with each other, and the first metal gate of one of the two first transistors is electrically connected with the second metal gate of one of the two second transistors.

15. The semiconductor device as claimed in claim 8, further comprising three first transistors connected with each other and three second transistors connected with each other, and the first metal gate of one of three first transistors is electrically connected with the second metal gate of one of the three second transistors.

16. The semiconductor device as claimed in claim 8, wherein the second metal gate further comprises a second P-metal layer stacked to the second N-metal layer; wherein the semiconductor device further comprises three first transistors connected with each other and three second transistors connected with each other, and the first metal gate of one of three first transistors is electrically connected with the second metal gate of one of the three second transistors.

17. The semiconductor device as claimed in claim 8, further comprises: a current mirror circuit electrically connected with the first transistor; a first resistor electrically connecting the current mirror circuit with a series voltage; and a second resistor electrically connecting the second transistor with the series voltage.

18. A semiconductor device, comprising: a first transistor, comprising: a plurality of first active channels stacked to each other; and a first metal gate formed on the first active channels, wherein the first metal gate comprises: a plurality of first P-metal layers; and a first N-metal layer stacked to the first P-metal layers; a plurality of second transistors, each comprising: a plurality of second active channels stacked to each other; and a second metal gate formed on the second active channels, wherein the second metal gate comprises a second N-metal layer; wherein one of the second transistors further comprises a second P-metal layer, and is electrically connected with another of the second transistors.

19. The semiconductor device as claimed in claim 18, wherein the first transistor has a first driving voltage threshold, and each second transistor has a second driving voltage threshold less than the first driving voltage threshold.

20. The semiconductor device as claimed in claim 18, further comprising two first transistors connected with each other, and the first metal gate of one of the two first transistors is electrically connected with the second metal gate of one of the two second transistors.
